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WORKSHEET 10

AIM: To design and implement a trigger in PostgreSQL that automatically generates primary key values, ensuring unique identification of records and demonstrating automated database programming.

S/W Requirement: **Database Management System:**

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

OBJECTIVES:

- To implement automatic primary key generation using triggers
- To eliminate manual key insertion errors
- To maintain data integrity using database automation
- To understand trigger execution in PostgreSQL

PROBLEM STATEMENT:

In enterprise systems, manually assigning primary keys can lead to:

- Duplicate values
- Data inconsistency
- Human errors

To overcome this, design a trigger that:

- Automatically assigns a unique primary key before inserting records
- Uses sequence-based logic

- Ensures consistency and reliability

This approach is widely used in companies like Amazon, Flipkart, and Oracle Corporation.

TABLE STRUCTURE:

```
DROP TABLE employee;

CREATE TABLE employee (
  emp_id INTEGER PRIMARY KEY,
  name VARCHAR(100),
  department VARCHAR(50)
);
```

```
CREATE SEQUENCE emp_seq START 1;
```

IMPLEMENTATION/CODE

PROGRAM:

```
CREATE OR REPLACE FUNCTION emp_trigger_function()
RETURNS TRIGGER AS $$
BEGIN
  NEW.emp_id := nextval('emp_seq');
  RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE FUNCTION

Query returned successfully in 121 msec.
```

```
CREATE TRIGGER emp_trigger
BEFORE INSERT ON employee
FOR EACH ROW
EXECUTE FUNCTION emp_trigger_function();
```

```
INSERT INTO employee (name, department) VALUES ('John', 'HR');
INSERT INTO employee (name, department) VALUES ('Alice', 'IT');
```

```
INSERT 0 1

Query returned successfully in 140 msec.
```

```
SELECT * FROM employee;
```

OUTPUT:

Data Output | Messages | Notifications

	emp_id [PK] integer	name character varying (100)	department character varying (50)
1	3	John	HR
2	4	Alice	IT

LEARNING OUTCOMES:

- Understand the concept of database triggers
- Implement automatic primary key generation
- Ensure data integrity without manual intervention
- Apply trigger-based automation in real-world systems
- Gain practical exposure to enterprise-level database techniques

CONCLUSION:

The experiment successfully demonstrated the use of triggers in PostgreSQL to automate primary key generation. By integrating sequences with triggers, the system ensures unique record identification without manual input. This method enhances data integrity, reduces errors, and aligns with industry practices followed by modern enterprise applications.